

Topology Optimization of the Stator for Minimizing Cogging Torque of IPM Motors

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This study presents a topology optimization method to reduce cogging torque, applied to the design of the stator core of an interior permanent magnet motor. The proposed method uses a reaction-diffusion equation and optimization is separately performed based on two objective functions: 1) minimization of the cogging torque itself, for a number of rotor positions; and 2) minimization of the magnetic energy variation. The optimization results show that configurations having dummy slots are desirable, but the number of dummy slots differs according to the objective function used.

Index Terms—Cogging torque, finite-element analysis, interior permanent magnet (IPM) motor, reaction-diffusion equation, topology optimization.

I. INTRODUCTION

INTERIOR permanent magnet (IPM) motors offer several advantages such as wide speed range, high efficiency, and high torque-to-volume ratio, including flux-weakening operation. Such motors are widely used not only in household applications but also hybrid electric vehicles and electric vehicles. However, IPM motors are subject to the effects of cogging torque, caused by interactions between the permanent magnets (PMs) embedded in the rotor core and stator core saliencies. This prevents smooth rotation of the rotor and causes undesirable acoustic noise and structural vibration in practical applications. Various techniques can reduce the cogging torque in PM motors, such as skewing [1], variation of PM spans in surface-mounted PM motors [2], and the use of notches in the stator teeth in order to obtain dummy slots [3].

Thus far, research pertaining to the structural optimization of motors based on design sensitivities has been historically quite rare. Size optimization is a traditional design method, but achieving optimal configurations beyond initially proposed designs using this method has been problematic. Quite recently, however, research on motor design using structural optimization methods has been presented, in which the design goal is minimization of torque ripple. Lee *et al.* [4] designed switched reluctance motors based on the density method. Kwack *et al.* [5] optimized the stator shape of IPM motors using a level set-based design method.

This study presents a topology optimization method applied to the design of IPM motor stator cores to reduce cogging torque using a reaction–diffusion (R–D) equation. For the stator design, two design goals are considered. First, we minimized the cogging torque itself. Second, we reduce the cogging torque by reducing the variation of the magnetic energy against rotor movement. The system response and the R–D equation are solved using the 2-D finite-element method. The augmented Lagrangian method is employed to deal with constraints, and

the design sensitivity is computed based on the adjoint variable method.

II. TOPOLOGY OPTIMIZATION USING AN R–D EQUATION

A. Approach Based on an R–D Equation

R–D systems are mathematical models that use partial differential equations composed of two parts: a local “reaction” part describing how substances react with each other, and a “diffusion” part that describes how the substances spread in space. These equations can model the propagation of the interface between two phases, which is useful when analyzing population dynamics [6], chemical reactions [7], and biological morphogenesis [6], [8]. Recently, structural optimization using R–D equations has been explored [9]–[11].

The topology optimization method proposed by Choi *et al.* [11] yields optimal configurations that are free from checkerboard patterns and show minimal mesh dependence. The material properties are determined based on the density field, which is the solution of the R–D equation expressed as follows:

$$\frac{\partial \phi(t)}{\partial t} = \alpha \nabla^2 \phi(t) - \frac{\partial \bar{f}(\phi)}{\partial \phi}, \quad \frac{\partial \phi(t)}{\partial \mathbf{n}} = 0 \quad (1)$$

where ϕ is the density field and the reaction term $\partial \bar{f}(\phi)/\partial \phi$ is the derivative of the augmented Lagrangian $\bar{f}$. The Neumann condition is applied over the entire boundary of the design domain Ω_D . α and $\mathbf{n}$ represent the diffusion coefficient and the outer unit vector normal to the domain boundary, respectively. In this study, the initial condition $\phi(0) = \phi_0 \in L^2(\Omega_D)$ with $0 < \phi_0 < 1$ is given and $\phi \in H^1(\Omega_D)$ is solved such that $\int_{\Omega_D} \phi dx / \int_{\Omega_D} dx = V_{\text{req}}$ and $0 \leq \phi \leq 1$, where V_{req} is the required volume fraction.

In our approach using the R–D equation, the structural boundary is represented by the diffuse-interfacial layer between the solid and the void domains as shown in Fig. 1. In this study, we define boundary Γ as

$$\Gamma(\mathbf{x}, t) = \{\mathbf{x} \in \Omega | \phi(\mathbf{x}, t) = 0.5\}. \quad (2)$$

The thickness of the interfacial layer is determined by the diffusion coefficient value α . Generally, as the diffusivity is reduced, the thickness is decreased, which causes the interface to become steeper.

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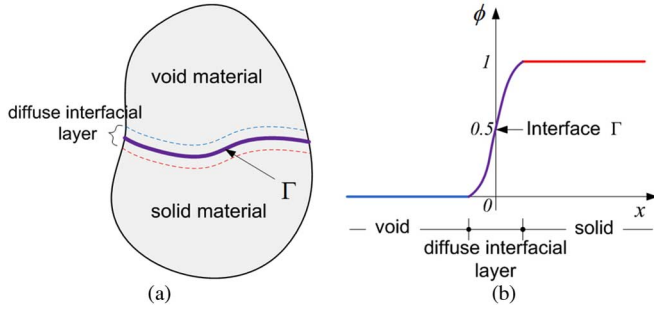


Fig. 1. Structural boundary representation: (a) separation of material phases using a diffuse interface and (b) cross section of the diffuse interface.

B. Augmented Lagrangian

Assuming that the objective function $f(\phi)$ is to be minimized, the current optimization problem can be written as

$$\begin{aligned} & \underset{\phi}{\text{minimize}} && f(\phi, A_z) \\ & \text{subject to} && h_1(\phi) = \int_{\Omega_D} \phi dx - V_{\text{req}} \int_{\Omega_D} dx = 0 \\ & && h_2(\phi) = \int_{\Omega_D} \phi(1 - \phi) dx - \varepsilon = 0 \\ & && 0 \leq \phi \leq 1 \end{aligned} \quad (3)$$

where A_z is the z -directional vector potential, i.e., the system response. The second constraint h_2 is used to avoid intermediate densities when $\varepsilon \approx 10^{-3} \times \int_{\Omega_D} dx$ has very small values. All constraints are implemented based on the augmented Lagrangian method [12]. This method is a combination of the Lagrange multiplier method and the penalty method, which converts constrained optimization problems to unconstrained problems. The augmented Lagrangian $\bar{f}$ is defined as

$$\bar{f}(\phi) = f(\phi) + \sum_{i=1}^2 \left[\lambda_i h_i(\phi) + \frac{r_i}{2} h_i(\phi)^2 \right] \quad (4)$$

where λ_i and r_i are the Lagrangian multiplier and the penalty parameter, respectively. They are simply updated using the following scheme:

$$\lambda_{i,k+1} = \lambda_{i,k} + r_{i,k} h_i(\phi), \quad r_{i,k+1} = \gamma r_{i,k}. \quad (5)$$

Here, k is the iteration number of the optimization process. γ is a fixed parameter for increasing the convergence rate.

The gradient of the augmented Lagrangian with respect to the density field is directly used as the reaction term in (1) and is defined as

$$\frac{\partial \bar{f}(\phi)}{\partial \phi} = \xi \frac{\partial f(\phi)}{\partial \phi} + \sum_{i=1}^2 \left[\frac{\partial h_i(\phi)}{\partial \phi} (\lambda_i + r_i h_i(\phi)) \right] \quad (6)$$

where ξ is a parameter normalizing the design sensitivity $\partial f / \partial \phi$ and is defined as

$$\xi = \frac{\int_{\Omega_D} dx}{\|\partial f(\phi) / \partial \phi\|_{L^2(\Omega_D)}}. \quad (7)$$

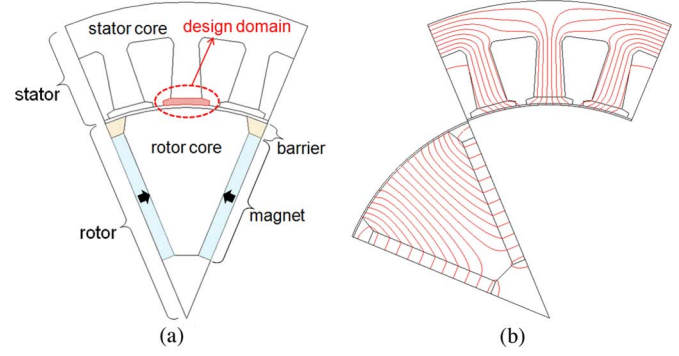


Fig. 2. IPM motor design model (air gap: 0.6 mm, stack length: 65 mm): (a) schematic diagram and (b) flux line at a rotor angle of 45°.

C. Optimization Procedure

The optimal configuration is obtained using the following procedure.

- 1) Initialize α , ϕ , λ_i , r_i , and γ .
- 2) Solve the system response A_z .
- 3) Compute the objective value $f(\phi)$ and the design sensitivity $\partial f(\phi) / \partial \phi$.
- 4) Update the density field ϕ using (1).
- 5) If not converged, repeat steps 2–4.

In this study, we use $\alpha = 10^{-4} \times \int_{\Omega_D} dx$, $\phi_0 = V_{\text{req}}$, $\lambda_{i,0} = 0$, $r_{i,0} = 0.02$, and $\gamma = 1.03$. In particular, parameter γ (> 1) accelerates the growth rate of the penalty parameter r_i and the convergence rate. However, convergence instability may result if r_i changes too rapidly.

For the optimization to minimize cogging torque, the design sensitivity is obtained using the adjoint variable method [12]. For our research, this method is more efficient compared with the direct method because the number of the design variables greatly exceeds the number of constraints in the topology optimization.

III. NUMERICAL EXAMPLES

The design model has three-phase winding, eight magnetic poles and 24 stator slots. Fig. 2 shows a 45° segment of the motor. The outer diameters of the rotor and the stator are 80 and 120 mm, respectively. The remanent flux density of the employed PM material is 1.1 T. Only one stator shoe is defined as the design domain, as shown in Fig. 2(a). At each iteration step, the density field distribution of this region is copied to the stator shoes on either side. The design domain and the entire domain are discretized using 1162 and 10 552 triangular three-node elements, respectively.

The evolution of the density field and the system analysis are achieved using Comsol Multiphysics, a commercial finite-element analysis package, and MATLAB. The use of the “moving mesh” application mode in Comsol Multiphysics supports analysis at various rotor positions without remeshing, as shown in Fig. 2(b).

For the topology optimization of the stator core, the magnetic reluctivity-based interpolation scheme [13] is used as specified in the following equation:

$$\nu(\phi, B^2) = \nu_0 + (\bar{\nu}(B^2) - \nu_0) \phi \quad (8)$$

where ν_0 and $\bar{\nu}$ are the reluctivity of free space and the iron employed, respectively. In this study, we consider both linear

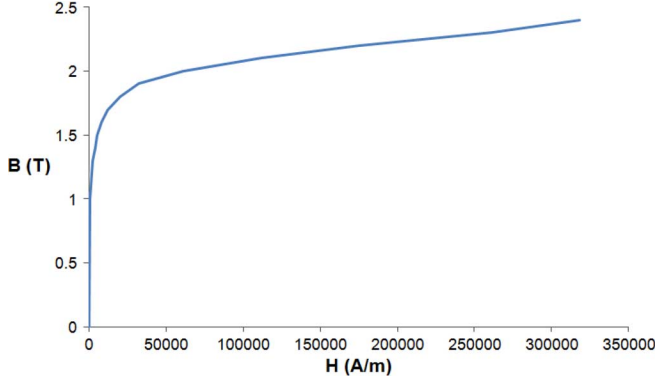


Fig. 3. Nonlinear B–H curve for soft iron.

TABLE I
OPTIMAL CONFIGURATION AND TORQUE RIPPLE BY TORQUE MINIMIZATION

		$V_{req} = 0.65$	$V_{req} = 0.8$
Linear case	Optimal shape		
	Torque ripple	0.281 (68.0%↓)	0.569 (35.1%↓)
Nonlinear case	Optimal shape		
	Torque ripple	0.654 (40.2%↓)	0.551 (49.6%↓)

and nonlinear permeability cases. In the case, considering the magnetic saturation, $\bar{\nu}$ is a function of the flux density, as shown in Fig. 3. In the linear case, a relative permeability of 1000 ($\bar{\nu} = 10^{-3} \times \nu_0$) is used.

A. Minimization of Cogging Torque

In order to reduce the cogging torque, we first consider a torque minimization problem where the torque is computed based on the Maxwell stress tensor. Here, the objective function is defined as

$$f(\phi, A_z) = \sum_{i=1}^N T_i^2 \quad (9)$$

where T_i is the cogging torque computed at the i th position. A mechanical angle of 0–45° is considered for the optimization. When the rotor is placed at the position shown in Fig. 2(a), the rotor angle is assumed to be 0°. We consider $N = 25$ different rotor positions as the rotor turns counterclockwise from 0° to 45°, increasing by 1.875° for each optimization iteration. The use of 25 rotor positions represents a good compromise between calculation time and detail needed to analyze the cogging torque profile.

Table I shows the optimal configurations of the stator shoes that minimize the cogging torque. All optimal stator shoes have two dummy slots regardless of the volume constraint or the analysis type, although the shapes are slightly different. The initial torque ripple values (peak-to-peak amplitude) for the linear and the nonlinear cases are 0.8775 and 1.094, respectively. Compared with the initial design model, the optimal models show a 35.1–68.0% decrease in torque ripple. The added dummy slots reduce the cogging torque and roughly triple its frequency, as shown in Fig. 4.

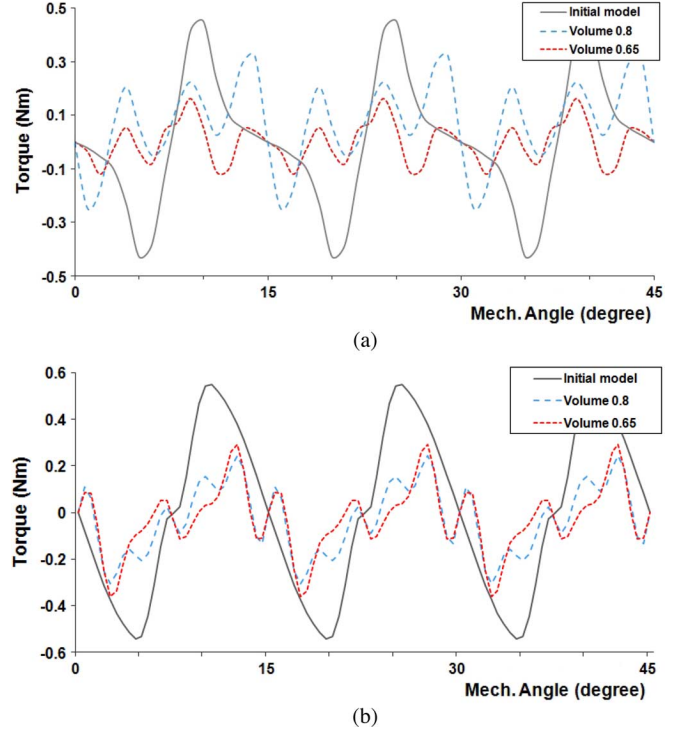
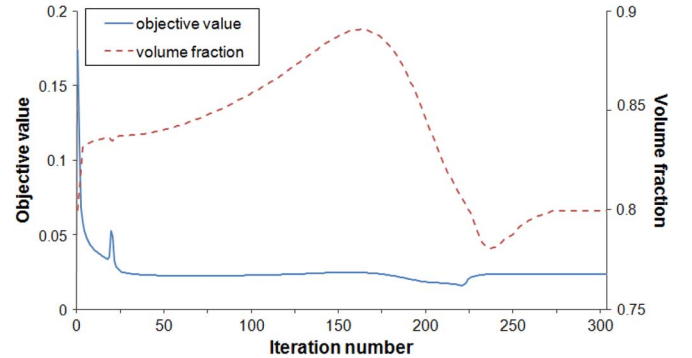
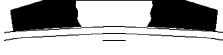


Fig. 4. Cogging torque profiles of optimal models by torque minimization: (a) linear models and (b) nonlinear models.

Fig. 5. Convergence history for the objective value and volume fraction (linear case, $V_{req} = 0.8$).

During the optimization process, we observe a slight change in density after the first iteration in areas that evolve to become dummy slots; however, about 300 iterations are required for convergence. The convergence history of the objective value and volume fraction for the linear case ($V_{req} = 0.8$) is somewhat unusual, as shown in Fig. 5. When the augmented Lagrangian method is used to deal with the volume constraint in the topology optimization, the volume convergence history typically shows an oscillation above and below the required volume fraction, resembling damped oscillation of a structural vibration. However, in our optimization problem, an additional constraint, h_2 in (3), was employed to avoid intermediate densities. As a result, during the optimization process, intermediate densities are under increasingly heavy pressure to converge to a value of 0 or 1. We note that this pressure has a deleterious effect of increasing the objective value, as shown in Fig. 5 after approximately iteration 220.

TABLE II
OPTIMIZATION RESULTS BY MINIMIZATION OF ENERGY VARIATION

	$V_{req} = 0.65$	$V_{req} = 0.8$
Optimal configuration		
Energy ripple	0.0291 (40.4%↓)	0.0107 (78.1%↓)
Torque ripple	1.115 (1.9%↑)	0.760 (30.6%↓)

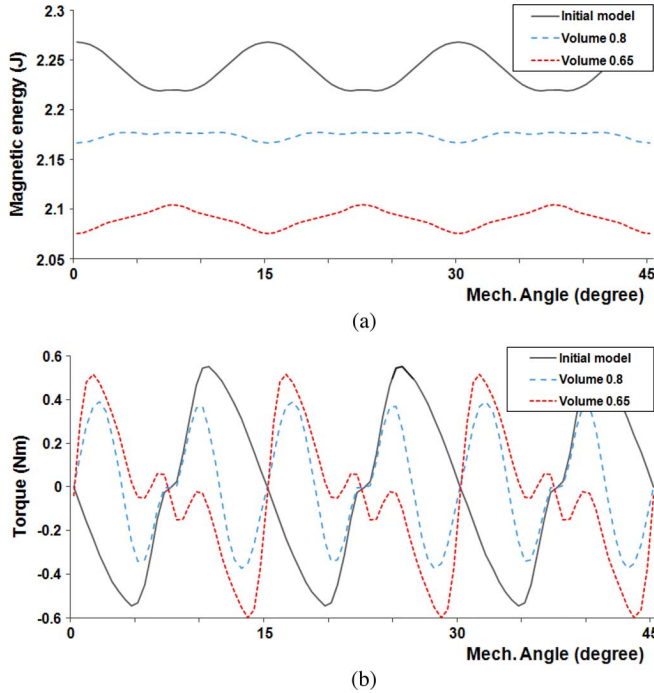


Fig. 6. Profiles of the magnetic energy and the cogging torque by minimization of energy variation: (a) magnetic energy and (b) cogging torque.

B. Minimization of Energy Variation

Cogging torque occurs due to the variation of the magnetic energy stored in the motor system when the rotor rotates. It can be reduced by minimizing the energy variation at various rotor positions [14]. Therefore, the objective function is now defined as

$$f(\phi, A_z) = \sum_{i=1}^N (W_i - W_{\text{avg}})^2 \quad (10)$$

where W_i is the magnetic energy at the i th position and W_{avg} represents the average energy level at the previous optimization iteration.

Table II shows the optimal stator shapes considering the magnetic saturation of soft iron. In contrast with the previous optimization problem results, only one dummy slot appears in the optimal configurations. As shown in Fig. 6, the energy ripple (peak-to-peak amplitude) is effectively reduced regardless of the required volume constraint. On the other hand, the cogging torque is not always reduced. With $V_{\text{req}} = 0.8$, the torque frequency is exactly doubled and the torque ripple falls by 30.6% compared with the initial model. However, when $V_{\text{req}} = 0.65$,

although the energy ripple is decreased, the torque ripple is increased beyond that of the initial model.

IV. CONCLUSION

In this study, a topology optimization method using an R-D equation was applied to the stator design of an IPM motor to reduce the cogging torque. The objective functions in two problems aimed to minimize the torque and the magnetic energy variation, respectively. In the optimal configurations, one or two dummy slots appeared, depending on the objective function used. In the optimization to minimize the energy variation, the energy ripple was decreased regardless of the volume constraint value, but the cogging torque was not always reduced. Therefore, we observe that torque minimization is a more appropriate objective function setting for reducing the cogging torque in IPM motors. The optimized models do not lead to deterioration in motor performances such as the torque ripple and the back-electromotive force.

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REFERENCES

- [1] N. Bianchi and S. Bolognani, "Design techniques for reducing the cogging torque in surface-mounted PM motors," *IEEE Trans. Ind. Appl.*, vol. 38, no. 5, pp. 1259–1265, Sep./Oct. 2002.
- [2] T. Li and G. Slemon, "Reducing of cogging torque in permanent magnet motors," *IEEE Trans. Magn.*, vol. 24, no. 10, pp. 2901–2903, Nov. 1988.
- [3] M. Goto and K. Kobayashi, "An analysis of the cogging torque of a DC motor and a new technique of reducing the cogging torque," *Elect. Eng. Jpn.*, vol. 103, no. 5, pp. 113–120, 1983.
- [4] J. Lee, J. H. Seo, and N. Kikuchi, "Topology optimization of switched reluctance motors for the desired torque profile," *Struct. Multidisc. Optim.*, vol. 42, pp. 783–796, 2010.
- [5] J. Kwack, S. Min, and J. Hong, "Optimal stator design of interior permanent magnet motor to reduce torque ripple using the level set method," *IEEE Trans. Magn.*, vol. 46, no. 6, pp. 2018–2111, Jun. 2010.
- [6] J. D. Murray, *Mathematical Biology*. Berlin, Germany: Springer-Verlag, 1993.
- [7] J. Ross, S. C. Müller, and C. Vidal, "Chemical waves," *Science*, vol. 240, pp. 460–465, 1988.
- [8] A. M. Turing, "The chemical basis of morphogenesis," *Philos. Trans. R. Soc. B-Biol. Sci.*, vol. 237, no. 641, pp. 37–72, 1952.
- [9] A. Takezawa, S. Nishiwaki, and M. Kitamura, "Shape and topology optimization based on the phase field method and sensitivity analysis," *J. Comput. Phys.*, vol. 229, pp. 2697–2718, 2010.
- [10] T. Yamada, K. Izui, S. Nishiwaki, and A. Takezawa, "A topology optimization method based on the level set method incorporating a fictitious interface energy," *Comput. Methods Appl. Mech. Eng.*, vol. 199, pp. 2876–2891, 2010.
- [11] J. S. Choi, T. Yamada, K. Izui, S. Nishiwaki, and J. Yoo, "Topology optimization using a reaction-diffusion equation," *Comput. Methods Appl. Mech. Eng.*, vol. 200, pp. 2407–2420, 2011.
- [12] A. D. Belegundu and T. R. Chandrupatla, *Optimization Concepts and Applications in Engineering*. Englewood Cliffs, NJ: Prentice-Hall, 1999.
- [13] J. S. Choi and J. Yoo, "Structural optimization of ferromagnetic materials based on the magnetic reluctivity for magnetic field problems," *Comput. Methods Appl. Mech. Eng.*, vol. 197, pp. 4193–4206, 2008.
- [14] D.-H. Kim, I.-H. Park, J.-H. Lee, and C.-E. Kim, "Optimal shape design of iron core to reduce cogging torque of IPM motor," *IEEE Trans. Magn.*, vol. 39, no. 3, pp. 1456–1459, May 2003.